Holonomy-Aware Topological AlphaZero: Self-Play Go on Non-Orientable and Arbitrary-Genus Boards

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Abstract

Game-playing agents assume flat rectangular boards: the convolutional towers of AlphaZero and KataGo are defined over $\mathbb{Z}^2$ tensors and cannot even ingest a board on a sphere, a torus, or a Möbius band. We introduce *Geodesics*, a topological generalization of Go to arbitrary closed and bordered surfaces (spheres, tori, cylinders, Möbius bands, Klein bottles, real projective planes) and lattices in one and three dimensions, under any of several mesh families — and ask what architecture plays it optimally. We present a first implementation of a *Holonomy-Aware Topological AlphaZero* (HATZ): an AlphaZero-style self-play agent whose network (i) carries the reflection ($\mathbb{Z}_2$) part of the gauge structure that non-orientable surfaces provably require, via an orientation cocycle computed from the board's combinatorics and orthogonal (Cayley-parameterized) copresheaf transport maps conditioned on it — *initialized at the mesh's geometric parallel transporter*, so the layer starts as a gauge convolution and self-play deforms it while orthogonality is guaranteed; (ii) is exactly equivariant to each board's automorphism group by construction on flat boards, replacing the dihedral data augmentation that has no analogue off the square grid; (iii) pools through interlevel-set components of its *own predicted ownership field* — a learned Morse function whose partition is shaped by supervision rather than differentiable relaxation — and *routes* message passing along that field's ascending manifolds, with a dedicated separatrix channel carrying the contested boundaries where Go fights live; (iv) laterally decomposes each position through an outcome-supervised edge filtration (a GIN-ε head predicting endpoint co-ownership) whose thresholds sit at *learned quantiles* of the current filter distribution (Hofer-style differentiable filtration: hard keeps are structure, soft gates carry gradient), with per-node attention across filtration levels, a successive-training curriculum over those levels, and the H_0 persistence pairs of the filtration injected as auxiliary inputs so the hierarchy carries every connection's topological lifetime; and (v) reads the board as a cell complex, passing messages over vertex–edge–face incidences with curvature and localized-homology inputs and an ownership auxiliary head. The network is small (4.8×10^4 parameters), dependency-free, and every gradient is hand-derived and numerically verified. We verify the orientation cocycle against ground-truth orientability on eleven board types, demonstrate machine-precision equivariance on flat boards, and report the pathologies specific to this setting — transposition-table ko cycles, a degenerate early-pass equilibrium under Tromp–Taylor scoring, and the failure of index-based tie-breaking (simulation of simplicity) to respect board symmetries when

*Draft v0.1. Code, boards, and the browser-playable implementation: https://samleventhal.com/subpages/geodesics/geodesics.html.

the Morse function is learned, an obstruction we show recurs at every level of the hierarchy (basins, persistence pairing, family merging) and resolve with value-and-set-defined constructions throughout — together with their remedies. To our knowledge this is the first game-playing agent on non-orientable boards; a full transfer and ablation study is specified and in progress.

1 Introduction

The rules of Go never needed a flat grid. Stones occupy vertices of a graph; liberties are empty neighbors; a chain lives while its liberty set is non-empty; territory is graph reachability. Everything downstream of these definitions — capture, ko, life and death, Tromp–Taylor scoring — is intrinsically combinatorial. The 19×19 lattice is a historical accident of the plane. *Geodesics* takes the rules at their word and instantiates them on discrete surfaces: geodesic and Goldberg spheres, square and hexagonal tori, Möbius bands, Klein bottles, the real projective plane, cylinders, and cubic lattices in E^3, with vertex, edge, face, or full cell-complex incidence structures.

This generality breaks every assumption baked into modern game networks. AlphaZero [1] and KataGo [2] use residual convolutional towers over rectangular tensors; their 8-fold dihedral data augmentation exploits a symmetry group that simply does not exist on a sphere (whose geodesic boards have icosahedral symmetry) or a Klein bottle (whose isometries include orientation-reversing translations). More fundamentally, Weiler et al. [13] prove that convolution on a non-orientable surface cannot reduce its structure group to $SO(2)$: any consistent feature transport must be steerable under reflections, because parallel transport around an orientation-reversing loop returns frames mirrored. A network that ignores this is discontinuous across the orientation seam of a Möbius or Klein board.

We ask: *what is the right architecture for optimal play across domain geometries and topologies*, and we build a first honest draft of the answer. Our contributions:

1. **The first game-playing agent on non-orientable and arbitrary-genus boards.** We compute a representative orientation cocycle $\varepsilon \in Z^1(G; \mathbb{Z}_2)$ of the first Stiefel–Whitney class from the board's combinatorics alone (dual-BFS over face cycles when faces exist; the quotient-gluing rule for fundamental-domain grids), verify it against ground truth on eleven board types, and pass messages through ε-conditioned transport maps so that crossing the seam applies a genuinely different learned map (§4.1).

2. **Exact automorphism equivariance without augmentation.** Weight-shared message passing over the board graph commutes with every graph automorphism; we keep all inputs Aut-invariant by construction and verify equivariance to machine precision, including under the seam-preserving isometries of the Möbius band (§4.8). This replaces dataset-specific augmentation with intrinsic symmetry, with the strict generalization benefit of equivariance [19].

3. **Ownership-driven topological pooling.** The scalar field whose level structure defines the pooling regions is the network's own mid-depth ownership prediction — a *learned, supervised* Morse function, following the hierarchical-segmentation program of [20]. Regions are the connected components of its interlevel sets (Black-leaning, contested, White-leaning), a position-dependent, parameter-independent operator through which backpropagation stays exact, and — unlike basin partitions with simulation-of-simplicity tie-breaking — one that is exactly automorphism-equivariant (§4.6).

4. **Outcome-supervised filtration levels.** Following the homophily-filtration methodology of the companion manuscript [18] and graph filtration learning [16], a GIN-ε edge head predicts endpoint *co-ownership* — Go heterophily is literal: contact fights are heterophilous edges — and fixed thresholds over it produce nested edge-filtered levels (chain cores → contested full graph). Message passing runs per level with learned per-node attention across levels, so gradients reach the filter through supervision and attention while the masks themselves stay combinatorial (§4.3).

5. **A cell-complex, ownership-supervised value/policy net** with curvature (angle-defect) and localized-homology inputs, KataGo-style global pooling, and an ownership head; 3.5×10^4 parameters, pure-`numpy` with hand-derived gradients verified numerically to 5×10^{-6} over every parameter (§4.5).

6. **Two self-play pathologies and their fixes**: PUCT descents that cycle forever through ko transpositions when the in-tree engine forgets history, cured by a positional-superko guard; and a degenerate early-pass equilibrium under Tromp–Taylor-plus-komi scoring that collapses the policy, cured by masking passes early in self-play (§5).

The literature intersection we occupy — game playing × gauge equivariance × non-orientable surfaces — is, to our knowledge, empty.

2 Related work

Game networks and size generality. KataGo trains one convolutional net across board sizes with masking and global-pooling bias layers and extrapolates beyond its training sizes [2]; Polygames obtains board-size invariance from fully convolutional trunks with global pooling [3]; ScalableAlphaZero replaces the CNN with a GIN and transfers from small to large Othello boards with an order-of-magnitude training-time reduction [4]. All operate on rectangular (or at least fixed-family) boards; none addresses topology.

GNN expressivity. Message passing is bounded by the 1-WL test [5, 6]; Go meshes are locally regular, which is precisely where 1-WL is weakest, motivating multi-aggregator updates [7], higher-order structure, and our cell-complex lifting, which is strictly more expressive than 1-WL [8].

Gauge equivariance. Gauge-equivariant CNNs on manifolds and meshes [11, 12] achieve equivariance to local frame choice via parallel transport; Weiler et al. [13] prove non-orientable surfaces force the structure group $O(2)$ (reflections included), demonstrating on a Möbius toy model. We implement the discrete reflection ($\mathbb{Z}_2$) part, which is the component non-orientability makes mandatory.

Topological deep learning. Simplicial and CW networks [8], combinatorial complex networks [9], and neural sheaf diffusion [10] pass messages over incidence structures with (in the sheaf case) learnable transport. Our boards *are* cell complexes, and our transport maps are a transfer-friendly sheaf: globally shared matrices conditioned on the edge's cocycle value, rather than per-edge free parameters.

Hierarchical pooling and topological segmentation. Graph U-Nets [14] and DiffPool [15] learn coarsenings; our pooling is the Morse–Smale multi-persistence hierarchy of a scalar field on the board, following the hierarchical GNN construction of [20], which gives a canonical segmentation into basins (territories) meeting along separatrices (contested frontiers).

3 Setting: Go on a surface

A *board* is a finite graph $G = (V, E)$, optionally equipped with a 2-cell structure (face cycles) making it a closed or bordered combinatorial surface, produced by `make_board(surface, mesh, resolution)`. The surfaces used here, with the smallest instances in our suite:

board	$\lvert V\rvert$	$\lvert E\rvert$	χ	orientable	seam edges
sphere (geodesic f=2)	42	120	2	yes	0
plane D^2 (5×5)	25	40	1	yes	0
cylinder (5×5)	25	45	0	yes	0
torus (5×5)	25	50	0	yes	0
Möbius (5×5)	25	45	0	no	5
Klein (5×5)	25	50	0	no	5
$\mathbb{RP}^2$ (5×5)	25	48	1	no	8
box E^3 (4^3)	64	144	—	—	0
3-torus (4^3)	64	192	—	—	0

Rules are graph-intrinsic (positional superko; Tromp–Taylor scoring with komi 7.5). Quotient boards are fundamental-domain grids with identifications; e.g. the Klein bottle identifies $(x+n_x, y) \sim (x, n_y-1-y)$ (a flip) and $(x, y+n_y) \sim (x, y)$; $\mathbb{RP}^2$ flips both pairs. The browser implementation and the training package construct adjacency independently and agree edge-for-edge (verified in tests), so trained agents transfer between them.

4 Architecture

4.1 Reflection gauge structure from combinatorics

A local orientation at a vertex is a $\mathbb{Z}_2$ gauge choice; an assignment $\varepsilon : E \to \{\pm 1\}$ records whether traversing an edge preserves or reverses it. Gauge transformations act by $\varepsilon'(u, v) = g(u)\, \varepsilon(u, v)\, g(v)$ for $g : V \to \{\pm 1\}$; the *holonomy* of a cycle, $\prod_{e \in \gamma} \varepsilon(e)$, is gauge-invariant, and the class $[\varepsilon] \in H^1(G; \mathbb{Z}_2)$ restricted to surface cycles represents the first Stiefel–Whitney class w_1: it is trivial iff the surface is orientable.

We compute a representative in two ways. When face cycles are available, a BFS over the dual graph assigns each face an orientation; a shared edge whose two incident faces traverse it incoherently under the assignment is a *seam* edge ($\varepsilon = -1$). For fundamental-domain grids the flipped gluing itself is the seam: x-wrap edges on Möbius and Klein boards, both wrap families on $\mathbb{RP}^2$.

Proposition 1. *The seam set produced by dual-BFS is a cocycle representing w_1; it is empty for some BFS iff the surface is orientable. On quotient grids, the flipped-gluing edge set is likewise a representative.*

Proof sketch. Dual-BFS fixes a gauge on faces along a spanning tree; conflicts can occur only on non-tree dual edges, and the conflict indicator is exactly the obstruction cocycle to extending the orientation, whose class is w_1. For quotient grids, cutting along the flipped gluing yields the orientable fundamental domain, so a loop reverses orientation iff it crosses the gluing an odd number of times. □

Empirically, the construction recovers ground-truth orientability on all eleven board types in our suite (including hexagonal-mesh Möbius bands and 3D lattices), with seam counts matching the gluing combinatorics (Table above).

4.2 Cayley-geometric copresheaf transport

Let $h_v \in \mathbb{R}^D$ be vertex features. Messages pass through restriction maps conditioned on the cocycle and composed with the mesh's own parallel transporter,

$$m_v \;=\; \tfrac{1}{|N(v)|} \sum_{u \in N(v)} \Big(W_0\, R(\theta_{v \leftarrow u}) + \varepsilon(u,v)\, W_1 \Big)\, h_u, \qquad W_0, W_1 \in O(D), \tag{1}$$

where $\theta_{v\leftarrow u}$ is the discrete Levi–Civita transport angle along the edge (computed from tangent frames on boards with an embedded mesh; identically 0 on flat boards) acting as 2×2 rotation blocks on channel pairs, and orthogonality of the learned maps is enforced by the Cayley parameterization $W = (I - \Lambda)(I + \Lambda)^{-1}$, $\Lambda = T - T^{\top}$ for a raw parameter T (backward pass derived in closed form and verified numerically). Because both ε and $(\cos\theta, \sin\theta)$ enter linearly, the layer still reduces to fixed aggregation matrices — a plain-cosine matrix, an antisymmetric sine matrix, and the ε-signed matrix — and remains trivially differentiable by hand. Crossing the orientation seam applies a different learned map — the mechanism by which neural sheaf diffusion [10] encodes orientation reversal and heterophily — but with W_0, W_1 *shared globally* rather than learned per edge, so the same parameters transfer across boards. On non-orientable boards the sine channel is zeroed *exactly*: w_1 obstructs a globally consistent rotation sign, and the seam crossing is carried by the reflection-aware ε path instead — the obstruction is represented, not papered over.

Remark 1 (Copresheaf transport versus gauge convolution). *Two literatures offer transport on surfaces.* Copresheaf/sheaf message passing *[10] attaches a learned linear map to every (directed) edge: maximally flexible — the maps can discover tactically meaningful directions from data — but per-edge parameters do not transfer across boards, nothing constrains the maps to preserve feature geometry, and on heterophilic graphs unconstrained maps are known to help precisely because they can do* anything. Gauge-equivariant convolution *[11, 12, 13] fixes the transport to the mesh's geometric parallel transporter: exactly norm-preserving, exactly determined by the geometry, zero parameters — but also zero capacity to learn that influence in Go propagates differently along a contested wall than across open territory. Eq. 1 is genuinely between the two. The raw Cayley parameter is initialized at $T = 0$, so $W_0 = I$ and the layer* starts as the gauge convolution's geometric transporter $R(\theta)$*; self-play then deforms W_0 away from the identity — a learned, globally shared correction that can encode game-specific transport — while the Cayley map keeps every intermediate point exactly orthogonal, preserving the gauge convolution's key virtue (norm preservation, hence genuine discrete parallel transport rather than arbitrary mixing) at every step of training. Geometric prior, learned correction, orthogonality guaranteed.*

Honest scope. The ε path is the discrete reflection part of the $O(2)$ structure non-orientability mandates [13]; the rotational part is now initialized from the mesh geometry on embedded boards but the *kernel* remains isotropic (full $SO(2)$ steerability is future work). The network consumes ε and the tangent frames in a fixed canonical gauge; holonomy-derived inputs are gauge-invariant, but the learned transport is gauge-covariant only up to reparameterization. We therefore train with *gauge augmentation* on both structures: each self-play game and each optimization step resamples a vertex sign field g and uses $\varepsilon'(u,v) = g(u)\varepsilon(u,v)g(v)$, and, on boards with nontrivial transport, resamples per-vertex frame rotations β and uses $\theta' = \theta + \beta_v - \beta_u$ — both leave holonomy (the gauge-invariant content) untouched while teaching the invariance the architecture does not yet guarantee.

4.3 Outcome-supervised filtration levels

Adjacent opposite-colored stones are heterophilous edges in the literal game-theoretic sense: contact fights concentrate heterophily, chains are homophilous cores. We port the homophily-based filtration methodology of [18] with class labels replaced by game outcomes. A GIN-ε head [5, 16] over the input embedding produces a symmetric per-edge filter $\phi(u,v) = \sigma(\mathrm{MLP}[g_u + g_v;\ |g_u - g_v|])$ supervised by *co-ownership*: whether the endpoints are owned by the same player under final Tromp–Taylor scoring (targets 1 / 0 / 0.5 for same/opposite/neutral).

Learned quantile thresholds. v2 used fixed thresholds $(0.75, 0.5, \text{full})$. But heterophily is not stationary within a game: early positions have few contact edges and a tight ϕ distribution, while a midgame contact fight spreads ϕ wide. Level ℓ therefore keeps edges with $\phi \geq p_\ell$ where p_ℓ is the *interpolated empirical quantile* of the current position's ϕ distribution at a learned fraction $\sigma(q_\ell)$ (initialized to 0.75 and 0.5), so the levels slide to track the actual contact structure: early game, levels cluster near the full graph; in a fight, they separate to isolate it. Differentiability follows the split introduced by Hofer et al. [16]: the hard keep set and its hard-count row normalization are structure (stop-grad), while a soft gate $\sigma\big((\phi_e - p_\ell)/\tau\big)$ multiplies each kept edge's aggregation weight, carrying gradient to ϕ_e, to the threshold p_ℓ, and — through the two bracketing order statistics of the interpolated quantile — to the sorted ϕ values and the learned fractions q_ℓ. No relaxation of the combinatorics is needed. The transport of Eq. 1 runs *per level* (the twisted aggregation matrices simply restrict to each kept, gated edge set, so gauge structure and equivariance are untouched), and levels combine by per-node attention $h_v = \sum_\ell \alpha_{v,\ell}\, m_v^{(\ell)}$, $\alpha_v = \mathrm{softmax}_\ell(m_v^{(\ell)} \cdot a_\ell)$, generalizing the filtration attention of [18]. Gradients reach the filter through its supervision, through the attention weights, and through the soft gates — always differentiable — while the level masks themselves are combinatorial structure: supervised, not relaxed, which sidesteps the piecewise-constant-membership problem entirely.

4.4 Persistence-pair injection

The levels of Sec. 4.3 sample the ϕ-filtration at a few learned heights; the filtration's H_0 *persistence diagram* carries strictly more — the topological lifetime of every connection. A descending Kruskal sweep over ϕ computes, per vertex, the (birth, death, lifetime, essential) of the superlevel component it was born into (birth: the vertex's first incident edge; death: absorption of its component by an elder one under the elder rule), and, per edge, its pairing role: a tree edge that killed a class (annotated with the lifetime it terminated) or a cycle edge whose ϕ is an H_1 birth. These features are injected additively into the vertex and edge embeddings through learned maps, so the hierarchy carries not just level membership but every connection's lifetime — a stone whose chain connection dies late under co-ownership is structurally different from one severed early, and the network now sees the difference. Pairing and values are structure (stop-grad); ϕ's gradient path remains its co-ownership supervision. Tie handling matters here too (Sec. 4.8): equal-ϕ edges are processed as simultaneous batches and equal-birth merges kill both sides, so the pairing depends only on values and sets, never on indices, and remains exactly automorphism-equivariant on the symmetric positions where a learned field produces exact ties.

4.5 Cell ranks and inputs

The board is read as a complex: vertex features always; edge features always (seam flag, endpoint color pattern); face features when face cycles exist. Per layer ℓ,

$$H_V \leftarrow \mathrm{relu}\big(H_V W_s + (AH_V)W_0 + (A_\varepsilon H_V)W_1 + (M_{EV}H_E)W_{ev} + \bar{h}\,W_g + b\big), \tag{2}$$

$$H_E \leftarrow \mathrm{relu}\big(H_E W_{es} + (M_{VE}H_V)W_{ve} + (M_{FE}H_F)W_{fe} + b_e\big), \qquad H_F \leftarrow \mathrm{relu}(\cdots), \tag{3}$$

with $M_{\cdot\cdot}$ fixed incidence-mean operators and $\bar{h}$ the global mean (KataGo-style pooling bias [2]). Inputs are Aut-invariant by construction: stone/empty indicators from the mover's perspective, normalized degree, discrete curvature (the Gauss–Bonnet angle defect $2\pi - \sum_i \theta_i$, whose total is $2\pi\chi$), localized H_0 data (own/opponent group size and liberty counts — the persistent-homology features in their cheapest useful form), and global scalars $[\,\mathbf{1}[w_1 \neq 0],\ \chi/|V|\,]$. Heads: a per-vertex policy logit plus a pooled pass logit; a pooled tanh value; and a per-vertex tanh *ownership* head supervised by final Tromp–Taylor territory, KataGo's auxiliary target adapted to graphs. The ownership head is applied twice with shared weights: at mid-depth (where its prediction becomes the pooling Morse function and the routing field of Sec. 4.7) and at the output, both supervised. Total: 47,783 parameters at width 32×3 layers — $\sim$190 KB, comfortably client-side. All gradients are hand-derived in `numpy`, including the Cayley inverse, the geometric rotation blocks, the per-node attention softmax, the soft-gated quantile filtration (through the gates, the thresholds, and the interpolated order statistics), and the filter head; central-difference verification over every parameter (with position structure frozen, since it is deliberately stop-grad) gives worst relative error 5.8×10^{-6} (Möbius), 9.4×10^{-7} (tetrahedron, live sine transport), and 4.4×10^{-6} (geodesic sphere), and $\|W^\top W - I\|_\infty < 10^{-12}$ for the transported maps, with $W_0 = I$ exact at initialization.

4.6 Ownership-driven topological pooling

The coarsening rung uses a *learned* Morse function: the mid-depth ownership prediction $\hat{o}$. Regions are the connected components of its interlevel sets $\{\hat{o} \geq t\}$, $\{|\hat{o}| < t\}$, $\{\hat{o} \leq -t\}$ at the median-$|\hat{o}|$ threshold — predicted Black territory, contested frontier, predicted White territory, the sublevel-set objects native to persistence. Features mean-pool to regions, message-pass once on the region graph, and unpool residually. The partition depends on the position and on $\hat{o}$ but is treated as stop-grad structure; the field is shaped by its own supervision ("supervised, not relaxed"), so backpropagation through the block is exact.

We initially used the ascending/descending *basin* partition of [20], and our equivariance tests caught a failure worth recording: with a learned field, symmetric positions produce value *plateaus*, and the index-based tie-breaking (simulation of simplicity) that discrete Morse theory uses for genericity then splits them arbitrarily — on the empty geodesic sphere the twelve equal pentavalent maxima carved the board into twelve index-dependent basins, and on the Möbius band two *chiral twin* seam-corner plateaus, exchanged by the band's reflection, merged through different union-find chains. Simultaneous merging of equal-persistence families repairs the first case but not the second. Interlevel-set components repair the class: components map to components under any symmetry of the field, so the partition is exactly automorphism-equivariant, verified to machine precision. Basin/separatrix structure is not abandoned — it returns, made equivariant, as the *routing* of message passing below.

4.7 Morse-flow directed propagation

The Morse–Smale segmentation of [20] entered v2 only as a pooling operator. Here it becomes the routing of message passing — the component most native to this line of work. Let f be the mid-depth ownership field: already computed, already supervised. Its discrete gradient orients edges (for $\{u, v\}$, direction is $\mathrm{sign}(f(v) - f(u))$, pointing uphill), and its ascending basins partition the board. Every vertex aggregation at layers past the mid-depth splits into three learned channels by an edge's role relative to the flow:

$$m_v = W_\uparrow \sum_{\substack{u \in N(v) \\ \text{up, same basin}}} \tfrac{h_u}{d_\uparrow(v)} + W_\downarrow \sum_{\substack{u \in N(v) \\ \text{down, same basin}}} \tfrac{h_u}{d_\downarrow(v)} + W_\leftrightarrow \sum_{\substack{u \in N(v) \\ \text{cross basin}}} \tfrac{h_u}{d_\leftrightarrow(v)}, \tag{4}$$

ascending (up-manifold), descending (down-manifold), and lateral/separatrix — edges between different basins, i.e. crossing a ridge where ascending flows diverge (in-basin plateau ties also route laterally: flat contested ground behaves as boundary). Isotropic message passing averages all three; Eq. 4 distinguishes them, and the separatrix channel makes contested boundaries — where Go fights live — first-class carriers of information rather than edges averaged away. Ascending manifolds are the paths along which influence flows toward each local maximum; they branch exactly at separatrices (the Morse–Smale cell boundaries). Routing messages along them means representations follow the influence field's actual branching structure instead of blending across it: a node near a contested boundary receives ascending messages from its own basin distinctly from lateral messages across the ridge, so it can represent "on the Black side of a fight with White pressing from *that* direction" — which mean aggregation cannot express.

Basin labels are computed by the same union-find machinery, with one equivariance-driven change (the chiral-plateau obstruction again): raw steepest-ascent pointers index-tie-break on plateaus, so instead each vertex carries the *set* of plateau-contracted maxima its ascending flows can reach — basin interiors coincide with steepest ascent, while boundary vertices are marked by reach-set divergence, which is precisely the Morse–Smale reading of a separatrix. The construction is value-and-set-defined throughout, hence exactly equivariant. Persistence simplification of the routing field is deliberately omitted: equal-persistence family merging cannot be made index-free (the same obstruction one level up), and over-segmentation only shifts edges into the lateral channel — exactly where contested structure belongs. As with the pooling partition, labels are stop-grad structure; f's gradient path is its own supervision.

4.8 Equivariance

Proposition 2. *If every input feature is invariant under* $\mathrm{Aut}(G)$ *(and, where used, under automorphisms of the complex and of* ε*), the network is exactly* Aut*-equivariant: for any automorphism* σ*,* $\pi(\sigma \cdot s) = \sigma \cdot \pi(s)$ *and* $v(\sigma \cdot s) = v(s)$*.*

Proof sketch. Each layer is a composition of permutation-equivariant operators ($A, A_\varepsilon, M_{\cdot\cdot}, S, P$ conjugate by the permutation; pointwise nonlinearities and global means commute with it) with shared weights. □

Verified to machine precision for the full v3 stack (gated quantile levels, attention, Cayley-geometric transport, persistence injection, Morse-flow routing, ownership-driven pooling): the empty-board policy is constant on the WL orbits of the plane and the torus, and exactly symmetric under the seam-preserving Möbius reflection — all spreads at floating-point noise. Exactness holds where the geometric transport is trivial ($\theta \equiv 0$: flat and, by the w_1 obstruction, non-orientable boards); on curved boards the tangent-frame choice is a gauge, and — exactly like the ε gauge — it is randomized during training rather than being exactly quotiented out.

Making the proposition true of the *whole* v3 stack required the chiral-plateau discipline at every structural decision, because a learned field produces exact value ties precisely on symmetric positions: the filter's structural decisions (ordering, keeps, persistence batching) are made on quantized values so floating-point noise cannot split an orbit; the persistence pairing processes equal-ϕ edges as simultaneous batches with set-defined deaths; the quantile keep sets are value comparisons; and the flow routing uses reach-sets instead of index-tie-broken pointers. Each of these replaced an index-dependent construction that our equivariance tests caught breaking symmetry — the same failure mode, recurring at every level of the hierarchy. On non-orientable boards the *gauge-dependent* remainder — seam placement, and frame choice on curved boards — is the component gauge augmentation trains away; we report this rather than hide it.

5 Self-play training, and two pathologies

Training is standard AlphaZero: PUCT (c=1.5, Dirichlet root noise), visit-count policy targets with temperature 1 for the first ten moves, replay buffer, Adam. The loss is $\mathrm{CE}(\pi, p) + (z - v)^2 + \frac{1}{2}\,\mathrm{MSE}(\hat{o}_{\mathrm{final}}) + \frac{1}{4}\,\mathrm{MSE}(\hat{o}_{\mathrm{mid}}) + \frac{1}{4}\,\mathrm{BCE}(\phi)$, all targets derived from final Tromp–Taylor territory. Gauge augmentation resamples the cocycle (and, on curved boards, the tangent frames) per game and per optimization step.

Successive-training curriculum over filtration levels. Following the successive-training result of [18] — heterophilous levels are where learning stalls, homophilous levels are where it starts — self-play and optimization can be restricted to the most homophilous prefix of the filtration levels, annealing one more level in every fixed number of iterations (the schedule position persists across warm-continued runs). Early iterations then train on the cleanest co-ownership level — separable chains, where a policy that has plateaued on the full contested graph can still make progress — before the contested levels are switched on. This directly targets the policy-plateau failure mode we observed on medium boards, and composes with the board-size curriculum (small-board pretraining, warm-continue to larger boards at higher simulation depth) already supported by the trainer. Two failures specific to this setting are worth recording.

Transposition ko cycles. Our search merges states in a transposition table keyed by (stones, mover, passes) and re-seats a scratch engine per step, which forgets ko history. Once two ko-mirror nodes exist with PUCT argmaxes pointing at each other, a descent cycles $A \to B \to A$ forever — the process hangs, typically from the second iteration once the tree is deep enough to contain a ko. The fix is the search-side analogue of the game's superko rule: any state repeated within a single descent scores as a draw and terminates (with a depth cap as backstop). Three hundred simulations through a live ko now complete in bounded time; the guard is a permanent regression test.

The early-pass equilibrium. Under Tromp–Taylor scoring with positive komi, an immediate double pass is a *guaranteed White win.* Shallow search discovers this instantly: White's visits collapse onto pass, half the replay buffer becomes one-hot pass targets, and the learned policy passes as both colors — losing every evaluation game to a random mover. (Loss decreases throughout; the curve looks healthy while the policy degenerates.) KataGo guards its self-play against early passing for the same reason [2]; we mask the pass action from search targets for the first $|V|/2$ moves, after which games run full length and evaluation strength reflects play rather than the scoring artifact.

6 Preliminary verification and evaluation protocol

What is verified now. Cocycle-vs-orientability on 11 board types; full-parameter gradient checks through every v3 path, including the live sine transport, the gated quantile filtration, and an odd-width unpaired transport channel (5.8×10^{-6} Möbius, 9.4×10^{-7} tetrahedron, 4.4×10^{-6} sphere); machine-precision equivariance as above; unit-modulus transport angles with $\theta \equiv 0$ exactly on flat and non-orientable boards; persistence-pair sanity (nonnegative lifetimes, elder-rule ordering, the $n-1$ / n_e-n+1 tree/cycle split); quantile-level nesting and gate bounds; exact three-way partition of directed edges by the flow router; curriculum masking and its schedule persistence across resumed runs; Gauss–Bonnet consistency of the curvature inputs (tetrahedron defect $= \pi$ per vertex); browser/Python adjacency parity edge-for-edge on cylinder, Klein, and $\mathbb{RP}^2$ boards; end-to-end training and serving (checkpoint $\rightarrow$ WebSocket bridge $\rightarrow$ browser opponent, including seam reconstruction on remote boards). Short v2 training runs on the non-orientable trio (Möbius, Klein, $\mathbb{RP}^2$; 24 simulations, six iterations) show healthy optimization of the five-term loss after the pass guard ($4.20 \rightarrow 3.39$; parity with a random opponent by iteration four) — reported as smoke evidence, not results.

Protocol (in progress). (i) *Zero/few-shot transfer*: train on k topologies, evaluate Elo on held-out ones (e.g. train sphere+torus, test Klein), against fixed anchors (random; the hand-designed Morse–Smale engine; a Stage-1 GNN of matched size). (ii) *Ablations*: remove in turn the ε-transport (W_1=0), the MSC block, the ownership head, curvature inputs, and gauge augmentation; measure Elo and gauge-sensitivity. (iii) *Ceiling*: on the plane boards only, compare against KataGo as an upper reference. (iv) *Symmetry*: measure sample efficiency versus an augmentation-based baseline, testing the equivariance dividend of [19] in self-play.

7 Limitations and future work

The transport is geometrically initialized but the kernel remains isotropic — full $SO(2)$ steerability over the rotation system is open; gauge invariance (seam placement and, on curved boards, frame choice) is learned, not guaranteed — a strictly equivariant formulation via the orientation double cover is the natural next step; persistent-homology inputs are the H_0 pairs of the single-parameter ϕ-filtration — multiparameter persistence over (co-ownership, influence) and explicit H_1 features for eyes/ko are pending; the persistence-unsimplified flow router trades noise robustness for exact equivariance, and a tie-free simplification scheme would recover both; automorphism groups are used for verification, not yet target symmetrization; and the empirical study above is the actual test of the thesis — in particular the curriculum's effect on the medium-board policy plateau. A browser export of the network is mechanical — the Morse–Smale machinery already ships in the client-side engine — and would make every trained agent playable with no installation.

8 Conclusion

Go's rules are indifferent to topology; the field's architectures are not. We built the first agent that plays where the boards get strange — carrying exactly the reflection structure that non-orientability forces, the automorphism symmetry each board actually has, and a topological pooling hierarchy with guarantees — small enough to ship in a browser tab, honest about which invariances are proved, learned, and pending.

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